

Mathematical Formulation of the Proactive Golden-Hour Bridge Model (PGBM) for Post-Earthquake UAV-Assisted Response

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Abstract—This document presents the formal mathematical formulation of the Proactive Golden-Hour Bridge Model (PGBM), a UAV-assisted emergency response system for post-earthquake scenarios. The system operates as an online decision-making framework where Responder UAVs are dynamically assigned to detected survivors to provide immediate assistance during the critical "Golden Hour" before ground rescue teams physically arrive. The model captures the coupling between survivor task states, UAV dynamics, battery constraints, time-dependent urgency decay, and 3D routing feasibility. The global system state is formally defined, and the event-triggered decision policy is mathematically characterized.

I. INTRODUCTION

In the aftermath of a major earthquake, Scout UAVs can rapidly detect trapped survivors using thermal and optical sensors. However, Ground Rescue Teams face significant delays due to collapsed infrastructure and blocked roads, creating a deadly gap between detection and physical rescue. The Proactive Golden-Hour Bridge Model (PGBM) introduces an intermediate UAV-assisted response layer where Responder UAVs are dispatched immediately to provide emergency supplies, two-way communication, and GPS relay before ground teams arrive.

A. System Overview

The PGBM consists of four primary actors:

- 1) **Scout UAVs:** Detect survivors and transmit detection packets.
- 2) **Command Center:** Maintains global state and executes online allocation/routing decisions.
- 3) **Responder UAVs:** Execute bridge missions to deliver aid.
- 4) **Ground Rescue Team:** Performs final physical extraction (outside core model).

B. Core Assumptions

- Obstacles are static during a single mission (Version 1).
- Communication between all actors is reliable.
- Each Responder UAV executes one tour at a time, serving up to K demand items (possibly multiple survivors, e.g., $K = 4$).

- Survivor tasks are independent except via shared tour capacity K , inventory I_j , and tour energy $E(\pi_j)$; each tour is non-preemptive (once dispatched, the order is fixed until return to base).

II. MATHEMATICAL NOTATION

Throughout this formulation, bold uppercase letters denote sets, bold lowercase denote vectors, and scalar values use standard notation. The time variable t denotes the current decision time (continuous). All decisions are event-triggered.

TABLE I
NOTATION SUMMARY

Symbol	Description
$\mathcal{S}(t), \mathcal{D}(t), \mathcal{E}$	Survivor set, Responder UAV set, Environment
$\mathcal{Q}_{\text{high}}, \mathcal{Q}_{\text{low}}$	High/low confidence queues ($c_i \geq c_{\text{thresh}}$)
$S_i = (p_i, t_i^{\text{det}}, c_i, \sigma_i, \mathcal{R}_i, a_i, \text{assigned_to}_i)$	Survivor task with demand set $\mathcal{R}_i$
$D_j = (p_j, p_j^{\text{base}}, B_j, K, I_j, m, A_j, s_j, \text{status}_j)$	Responder UAV with capacity K , inventory I_j , payload m
$p_i, p_j, p_j^{\text{base}} \in \mathbb{R}^3$	Positions and base station
$\sigma_i, c_i \in [0, 1], \mathcal{R}_i$	Severity, confidence, demand set $\subseteq \{\text{med, water, food}\}$
$u_i(t) = \sigma_i e^{-\alpha(t-t_i^{\text{det}})}$	Derived urgency, $\alpha = \ln 2 / T_{\text{half}}$
$B_j(t) \in [0, B_{\text{max}}], B_{\text{max}}, E_{\text{reserve}}$	Battery, max, reserve $0.2B_{\text{max}}$
$A_j \in \{0, 1\}, \text{status}_j, s_j$	Availability (Idle), status, current tour π_j
$T(\pi_j), E(\pi_j), C(\pi_j)$	Tour time, energy, cost (primary); T_{ij}, E_{ij}, C_{ij} is $k = 1$ special case
$T_{\text{max}}, E_{\text{max}}, c_{\text{thresh}}, \alpha, T_{\text{half}}, K$	Normalization, threshold, half-life, capacity
π_j, P_{π_j}, x_{ij}	Tour $(i_1, \dots, i_k)$, tour route, single-leg assignment ($k = 1$)
$X(t) = (\mathcal{S}, \mathcal{D}, \mathcal{E}, \mathcal{Q})$	Global state ($\mathcal{Q} = \mathcal{Q}_{\text{high}} \cup \mathcal{Q}_{\text{low}}$)

III. SYSTEM ENTITIES AND STATE DEFINITIONS

A. Survivor Task Model

Each survivor detection is modeled as a task $i \in \mathcal{S}(t)$, where $\mathcal{S}(t)$ is the set of all known survivor tasks at time t . Survivor i is defined as:

$$S_i = (x_i, y_i, z_i, t_i^{\text{det}}, c_i, \sigma_i, r_i, a_i, \text{assigned_to}_i) \quad (1)$$

with the following parameters:

TABLE II
SURVIVOR TASK PARAMETERS

Parameter	Symbol	Domain
Location	(x_i, y_i, z_i)	$\mathbb{R}^3$
Detection Time	t_i^{det}	$\mathbb{R}^+$
Confidence Score	c_i	$[0, 1]$
Severity	σ_i	$[0, 1]$
Demand Vector	r_i^q	$\mathbb{Z}_{\geq 0}, q \in \{\text{med, water, food}\}$
Assignment Flag	a_i	$\{0, 1\}$
Assigned UAV	assigned_to_i	$\mathcal{D}(t) \cup \{\emptyset\}$

r_i^q is the number of items of type q required by i (e.g., $r_i^{\text{med}} = 2, r_i^{\text{water}} = 0$); $|\mathcal{R}_i| = \sum_q r_i^q$ and $r_i^q > 1$ denotes multiple demands of same type at the same location. Severity σ_i denotes the static medical criticality at detection (independent of time), with $\sigma_i = 1$ representing the most critical condition. Two possible scenarios for obtaining σ_i are: (1) *Scout AI Inference*: estimated automatically from Scout UAV thermal/optical imagery and rubble context via a classification model (e.g., motion/heat signature analysis), and (2) *Human-in-the-Loop*: tagged manually by the Command Center operator reviewing the Scout video feed.

B. Urgency Decay Function

The urgency of survivor i decays exponentially with waiting time according to the Golden Hour principle:

$$u_i(t) = \sigma_i \cdot e^{-\alpha(t-t_i^{\text{det}})} \quad (2)$$

where $\sigma_i \in [0, 1]$ is the severity at detection so that $u_i(t_i^{\text{det}}) = \sigma_i$, $\Delta t = t - t_i^{\text{det}} \geq 0$ is the waiting time, and $\alpha > 0$ is the decay rate. The rate is defined via the half-life T_{half} (e.g., $T_{\text{half}} = 30$ min) as

$$\alpha = \frac{\ln 2}{T_{\text{half}}}, \quad (3)$$

so that $u_i(t_i^{\text{det}} + T_{\text{half}}) = \sigma_i/2$. For $T_{\text{half}} = 30$ min, $\alpha \approx 0.023 \text{ min}^{-1}$ ($3.8 \times 10^{-4} \text{ s}^{-1}$); $T_{\text{half}} = 60$ min gives $\alpha \approx 0.0115 \text{ min}^{-1}$. A survivor found 10 min ago ($\Delta t = 10$) has $u \approx 0.79\sigma_i$, while one found 90 min ago has $u \approx 0.12\sigma_i$ at the same current time t , capturing higher salvageability of fresher detections. The exponential form is chosen because it models proportional decay $du/dt = -\alpha u$ (a constant fraction of current urgency lost per minute), ensures $u(t) \in [0, 1]$ and never negative (unlike linear decay), and provides a single

interpretable parameter T_{half} ; a logistic threshold variant is deferred to future work for injury-specific cliff modeling.

C. Confidence Score and Verification Policy

Each detection carries a confidence score $c_i \in [0, 1]$ representing the Scout UAV detector's estimated probability that the detection is a true survivor (e.g., thermal/optical AI output). A threshold $c_{\text{thresh}} \in [0, 1]$ (e.g., 0.5) separates high- and low-confidence regimes:

$$\text{eligible}(i) = \begin{cases} 1 & \text{if } c_i \geq c_{\text{thresh}} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

Detections with $c_i \geq c_{\text{thresh}}$ enter the high-confidence queue $\mathcal{Q}_{\text{high}}(t)$ and are eligible for Responder assignment. Detections with $c_i < c_{\text{thresh}}$ enter the low-confidence queue $\mathcal{Q}_{\text{low}}(t)$ and are *not* eligible for Responder dispatch; they require verification (Scout re-scan or operator review) before promotion to $\mathcal{Q}_{\text{high}}$. In Version 1, $\mathcal{Q}_{\text{low}}$ is modeled as a holding set and verification is abstracted as an exogenous promotion event. The offloading ILP is defined only over $\mathcal{Q}_{\text{high}}(t)$.

D. Responder UAV Model

Each Responder UAV $j \in \mathcal{D}(t)$ is defined as:

$$D_j = (x_j, y_j, z_j, p_j^{\text{base}}, B_j, K, I_j^q, m, A_j, s_j, \text{status}_j) \quad (5)$$

with the following parameters:

TABLE III
RESPONDER UAV PARAMETERS

Parameter	Symbol	Domain
Position	(x_j, y_j, z_j)	$\mathbb{R}^3$
Base Station	p_j^{base}	$\mathbb{R}^3$
Remaining Battery	B_j	$\mathbb{R}^+$
Capacity	K	$\mathbb{N}$ (e.g., 4 items)
Inventory	I_j^q	$\mathbb{Z}_{\geq 0}, q \in \{\text{med, water, food}\}$
Payload Mass	$m_j(t) = \sum_q I_j^q(t)w_q$	$\mathbb{R}^+$ (kg, derived)
Availability	A_j	$\{0, 1\}$
Current Task	s_j	$\mathcal{S}(t) \cup \{\emptyset\}$
Mission Status	status_j	$\{\text{Idle, EnRoute, AtSurvivor, Returning}\}$

K is the maximum number of items and I_j^q is the inventory count of type q (e.g., $I_j^{\text{med}} = 2$). Payload $m_j(t) = \sum_q I_j^q(t)w_q$ is derived (w_q weight of type q) and decreases as I_j^q is consumed, affecting e_{travel} per leg.

E. Availability States

The availability A_j is derived from the UAV's status:

$$A_j = \begin{cases} 1 & \text{if } \text{status}_j = \text{Idle} \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

F. Environment Model

The disaster environment is represented as continuous sets:

$$\mathcal{E} = (V, O) \quad (7)$$

where $V \subset \mathbb{R}^3$ is navigable free space and $O \subset \mathbb{R}^3$ is the static obstacle set (buildings, rubble, no-fly zones), with $V \cap O = \emptyset$. For Version 1, $O(t) = O$ (static) and the hazard map $R : V \rightarrow [0, 1]$ is set to $R = 0$ and deferred. A route P is feasible iff $P \subset V$ (all waypoints lie in free space).

G. Task Queue

The ordered queue of unassigned survivors is defined as a derived view:

$$\mathcal{Q}(t) = \text{order}(\{S_i \in \mathcal{S}(t) \mid a_i = 0\}) \quad (8)$$

ordered by priority:

$$V_i(t) = u_i(t)c_i, \quad \text{priority}(i) = \beta V_i(t) \quad (9)$$

where $V_i(t) \in [0, 1]$ is the unified expected service value. Note: $\mathcal{Q}(t)$ is not an independent state component but a derived view of $\mathcal{S}(t)$ for scheduling. With verification threshold c_{thresh} , the queue partitions as:

$$\mathcal{Q}_{\text{high}}(t) = \{S_i \in \mathcal{Q}(t) \mid c_i \geq c_{\text{thresh}}\}, \quad (10)$$

$$\mathcal{Q}_{\text{low}}(t) = \{S_i \in \mathcal{Q}(t) \mid c_i < c_{\text{thresh}}\} \quad (11)$$

Only $\mathcal{Q}_{\text{high}}(t)$ is eligible for the offloading assignment; $\mathcal{Q}_{\text{low}}(t)$ is held for verification. Here c_{thresh} is a fixed policy parameter in V1 (e.g., 0.5), not an optimization variable, and $\mathcal{Q}_{\text{low}} \rightarrow \mathcal{Q}_{\text{high}}$ promotion is an exogenous verification event. *Uncertainty handling*: The partition manages detection uncertainty (false positives from Scout thermal/optical AI). High-confidence detections are directly eligible, while low-confidence detections are buffered in $\mathcal{Q}_{\text{low}}$ for verification (re-scan/operator review) before promotion, preventing scarce Responder UAVs from being wasted on likely false alarms. Note: $\text{priority}(i)$ is a pre-filter ordering heuristic for $\mathcal{Q}_{\text{high}}(t)$ (e.g., for greedy fallback or tie-breaking); the optimal assignment is determined by the ILP cost $C_{ij}(t)$, not by priority order.

IV. DECISION VARIABLES

A. Tour Variable (Primary Decision)

At any decision time t , the offloading decision for each available UAV j is an ordered tour

$$\pi_j(t) = (i_1, i_2, \dots, i_k), \quad k = \text{len}(\pi_j), \quad \sum_{i \in \pi_j} |\mathcal{R}_i| \leq K \quad (12)$$

where $i_l \in \mathcal{Q}_{\text{high}}(t)$ is the l -th survivor to be served, $|\mathcal{R}_i|$ is the number of items demanded by i , and K (e.g., 4) is the UAV item capacity. Thus π_j encodes both *selection* (which survivors) and *order* (in which sequence). For example, $\pi_j = (A, C)$ with

$\mathcal{R}_A = \{\text{med}\}$ and $\mathcal{R}_C = \{\text{water}, \text{food}\}$ carries $|\pi_j| = 3 \leq 4$ items to two survivors; $\pi_j = (C, A)$ is a different tour with different travel distance, climb, and payload-dependent energy. An empty tour $\pi_j = \emptyset$ means the UAV stays idle at base. The selected tours $\{\pi_j\}$ form a disjoint partial assignment over $\mathcal{Q}_{\text{high}}$ (each survivor in at most one tour; some may remain unassigned).

B. MILP Binary Variables

For MILP linearization, define $x_{ij}(t) \in \{0, 1\}$ with $x_{ij} = 1$ iff $i \in \pi_j(t)$, and $y_{jil}(t) \in \{0, 1\}$ with $y_{jil} = 1$ iff i is the l -th stop of j ($l = 1, \dots, K$). Then $\sum_l y_{jil} = x_{ij}$ and $\text{len}(\pi_j) = \sum_{i,l} y_{jil}$.

C. Tour Route (Geometric Realization)

A tour π_j is realized geometrically as the concatenation of its $k + 1$ optimal leg paths:

$$P_{\pi_j} = P_{p_j \rightarrow p_{i_1}}^* \circ P_{p_{i_1} \rightarrow p_{i_2}}^* \circ \dots \circ P_{p_{i_k} \rightarrow p_j^{\text{base}}}^* \quad (13)$$

where each leg $P_{a \rightarrow b}^* = \{a = p_0, p_1, \dots, p_L = b\}$ is the cheapest feasible waypoint sequence with $p_k \in V$. If any leg has no feasible path in V , the whole tour is infeasible. The routing oracle provides each leg: $\text{RoutingOracle}(p_a, p_b) \rightarrow (T_{ab}, E_{ab})$ or (∞, ∞) if infeasible.

V. COST FUNCTIONS

A. Path Cost and Optimal Path

For any path $P = \{p_0, \dots, p_L\} \subset V$, the traversal cost is defined as its energy:

For any path $P = \{p_0, \dots, p_L\} \subset V$, let $T(P) = \sum_k \|p_k - p_{k-1}\|/v_{\text{max}}$ and

$$E(P) = \sum_{k=1}^L [e_{\text{travel}}(p_k, m, w_0) \|p_k - p_{k-1}\| + e_{\text{vert}}(m) \max(0, z_k - z_{k-1})] \quad (14)$$

be the travel time and energy. The routing-layer objective is

$$J_{\text{route}}(P) = w_T \frac{T(P)}{T_{\text{max}}} + w_E \frac{E(P)}{E_{\text{max}}} \quad (15)$$

with $w_T, w_E > 0$. The cheapest feasible path is

$$P_{a \rightarrow b}^* = \arg \min_{P \subset V, p_0=a, p_L=b} J_{\text{route}}(P) \quad (16)$$

If no $P \subset V$ exists, the path is infeasible. The routing oracle encapsulates this: $\text{RoutingOracle}(a, b)$ returns (T_{ab}, E_{ab}) for $P_{a \rightarrow b}^*$ or (∞, ∞) if infeasible.

B. Tour Time and Energy

For a tour $\pi_j = (i_1, \dots, i_k)$, let $P_0 = P_{p_j \rightarrow p_{i_1}}^*$, $P_l = P_{p_{i_l} \rightarrow p_{i_{l+1}}}^*$ ($1 \leq l < k$), and $P_k = P_{p_{i_k} \rightarrow p_j^{\text{base}}}^*$. With $v_{\max}$ the maximum speed and $T_{\text{hover}} = T_{\text{drop}} + T_{\text{verify}} = 15$ s per stop,

$$T(\pi_j) = \sum_{l=0}^k \frac{\|P_l\|}{v_{\max}} + \sum_{i \in \pi_j} T_{\text{hover}}(i) \quad (17)$$

Let $m_0 = \sum_{\text{item} \in I_j} \text{weight}(\text{item})$ and $m_l = m_{l-1} - \sum_{\text{item} \in \mathcal{R}_{i_l}} \text{weight}(\text{item})$ be the payload before leg l . Then

$$E_{\text{leg}}(l) = e_{\text{travel}}(p_l, m_{l-1}, w_0) \|P_l\| + e_{\text{vert}}(m_{l-1}) \Delta z_l \quad (18)$$

with $e_{\text{travel}}(p, m, w_0) = e_0 + k_m m + k_w w_0$, $e_{\text{vert}}(m) = mg/\eta_{\text{vert}}$ (m, w_0 realtime; w_0 uniform wind, $w(p)$ deferred; $e_0, k_m, k_w, \eta_{\text{vert}}$ calibrated offline). The total tour cost and energy are

$$E(\pi_j) = \sum_{l=0}^k E_{\text{leg}}(l) + \sum_{i \in \pi_j} e_{\text{hover}} T_{\text{hover}}(i) \quad (19)$$

Early legs carry larger m and thus higher e_{travel} . The single-leg case $k = 1$ gives $E_{ij}^{\text{total}} = E(\pi_j = (i))$.

C. Coupling to Offloading

Routing provides $(T(\pi_j), E(\pi_j))$ via the oracle for each candidate tour. Offloading uses them in the feasibility check (21) and tour cost (??). All $k + 1$ legs must be feasible; otherwise $J_{\text{tour}}(\pi_j) = C(\pi_j) = \infty$ (Fig. 1). Here $J_{\text{tour}}(\pi_j) = C(\pi_j)$ and $J_{\text{route}}(P)$ is defined in (15). Note: Exhaustive enumeration $\sum_{k=1}^K P(N, k)$ is exponential (e.g., $N = 20, K = 4 \Rightarrow 123\,520$ tours per UAV); practical implementation prunes candidates via urgency-ranked beam search or spatial clustering to $B \approx 100$ tours before querying the oracle. The formulation defines the full set for completeness.

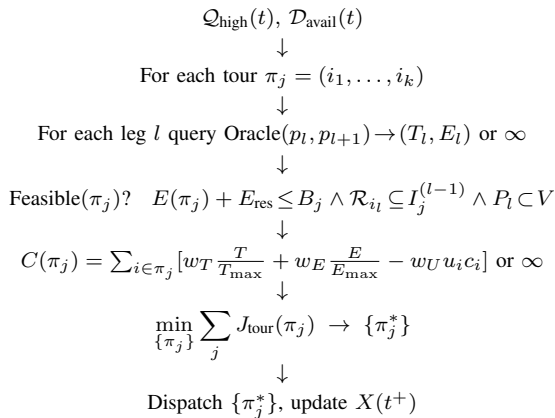


Fig. 1. PGBM coupling flow: Offloading queries Routing oracle per leg, aggregates $T(\pi_j), E(\pi_j)$, checks feasibility, computes $J_{\text{tour}}(\pi_j) = C(\pi_j)$, solves tour, and updates state.

D. Tour Assignment Cost

$$J_{\text{tour}}(\pi_j, t) = C(\pi_j, t) = \begin{cases} w_T \frac{T(\pi_j)}{T_{\max}} + w_E \frac{E(\pi_j)}{E_{\max}} - w_U \frac{\sum_{i \in \pi_j} V_i(t)}{K} \\ \infty \end{cases} \quad \text{if Feasible} \quad (20)$$

where $w_T, w_E, w_U > 0$ are tour cost weights ($w_T + w_E + w_U = 1$ baseline), $T_{\max} = 600$ s (fixed) and $E_{\max} = B_{\max} - E_{\text{reserve}}$ (40000 J) are fixed global constants, $V_i(t) = u_i(t)c_i$ is the unified service value, and $-w_U V_i$ rewards expected urgency per survivor. Infeasible tours have ∞ cost. The single-pair C_{ij} is $k = 1$ where $\pi_j = (i)$.

Rationale: The tour cost is a dimensionless trade-off among time, energy, and expected value summed over the tour; C_{ij} is $k = 1$.

VI. FEASIBILITY CONSTRAINTS

A. Battery Constraint for Tour (Big-M)

A tour π_j is feasible only if

$$\sum_{l,i} y_{jil} E_{\text{leg}}(l, i) + E_{\text{reserve}} \leq B_j(t) + M(1 - z_j) \quad (21)$$

where $E_{\text{leg}}(l, i)$ is precomputed via oracle for position l , $M = B_{\max}$ is big-M, and $z_j \in \{0, 1\}$ with $z_j = 1$ iff π_j feasible; $E(\pi_j) = \sum_{l,i} y_{jil} E_{\text{leg}}(l, i) + \sum_{i,l} y_{jil} E_{\text{hover}}$. If $z_j = 0$, $C(\pi_j) = \infty$.

B. Demand Matching

Each demand must be satisfied per type with explicit consumption:

$$I_j^{q,(l)} = I_j^{q,(l-1)} - r_{i_l}^q, \quad I_j^{q,(0)} = I_j^q, \quad I_j^{q,(l)} \geq 0 \quad \forall q, \forall l \quad (22)$$

Equivalently $r_{i_l}^q \leq I_j^{q,(l-1)}$ for all q and $i_l \in \pi_j$; if $r_{i_l}^q > I_j^{q,(l-1)}$ the tour is infeasible. This sequential check is stronger than the aggregated $\sum |\mathcal{R}_i| \leq K$.

C. Reachability per Leg

Each leg must lie in free space: $P_l \subset V$ for all l , i.e., $p_k \in V$ for every waypoint. Reachability is an oracle per leg; one infeasible leg makes the whole tour infeasible.

D. Capacity and Availability

$\sum_{i \in \pi_j} |\mathcal{R}_i| \leq K$ (total items) and $A_j = 1$ (Idle). If $A_j = 0$ or capacity exceeded, the tour is infeasible. Here $\text{len}(\pi_j) = k$ is the number of survivors and $\sum |\mathcal{R}_i|$ is the number of items.

E. Feasible Tour Condition

A tour π_j is feasible iff all of the above hold:

$$\begin{aligned} \text{Feasible}(\pi_j) = & (A_j = 1) \wedge \left(\sum_{i \in \pi_j} |\mathcal{R}_i| \leq K \right) \wedge (\forall i \in \pi_j : c_i \geq c_{\text{thresh}}) \\ & \wedge (\forall l : P_l \subset V) \wedge (\forall i_l \in \pi_j : \mathcal{R}_{i_l} \subseteq I_j^{(l-1)}) \\ & \wedge (E(\pi_j) + E_{\text{reserve}} \leq B_j(t)) \end{aligned} \quad (23)$$

VII. ASSIGNMENT CONSTRAINTS

A. Single Assignment per Survivor

Each survivor in $\mathcal{Q}_{\text{high}}(t)$ appears in at most one tour:

$$\sum_j \mathbf{1}\{i \in \pi_j(t)\} \leq 1 \quad \forall i \in \mathcal{Q}_{\text{high}}(t) \quad (24)$$

B. Capacity per Tour

Each tour respects item capacity:

$$\sum_{i \in \mathcal{Q}_{\text{high}}(t)} \sum_{l=1}^K \sum_q r_i^q y_{jil}(t) \leq K \quad \forall j \in \mathcal{D}_{\text{avail}}(t) \quad (25)$$

where K is the item capacity (e.g., 4). Equivalently $\sum_{i \in \pi_j} |\mathcal{R}_i| \leq K$. A UAV may stay idle ($\pi_j = \emptyset$, all $y_{jil} = 0$) if no feasible tour exists.

C. MILP Ordering Constraints

For $y_{jil}(t) \in \{0, 1\}$ ($l = 1, \dots, K$):

$$\sum_i y_{jil}(t) \leq 1 \quad \forall j, l, \quad \sum_l y_{jil}(t) = x_{ij}(t) \quad \forall i, j, \quad \sum_j x_{ij}(t) \leq 1 \quad \forall i \quad (26)$$

and $\sum_i y_{jil} \geq \sum_i y_{j, i, l+1}$ (no gap: position $l+1$ occupied only if l occupied).

D. Feasibility Enforcement

If a tour π_j is not feasible, its cost is set to infinity: $C(\pi_j) = \infty$ when $\neg \text{Feasible}(\pi_j)$ (includes $c_i < c_{\text{thresh}}$, $A_j = 0$, $P_l \not\subset V$, $\mathcal{R}_{i_l} \not\subseteq I_j^{(l-1)}$, or $E(\pi_j) + E_{\text{reserve}} > B_j$). Any optimal solution will have such tours excluded. In MILP form, $C(\pi_j) = \sum_{i,l} y_{jil} c_{jil}$ with $c_{jil} = w_T T_{\text{leg}}(l, i) / T_{\text{max}} + w_E E_{\text{leg}}(l, i) / E_{\text{max}} - w_U u_i c_i$.

VIII. OBJECTIVE FUNCTION

A. Primary Objective: Minimize Cost

The snapshot MILP over y_{jil} is:

$$\min_{x_{ij}, y_{jil}} \sum_{j \in \mathcal{D}_{\text{avail}}} \sum_{l=1}^K \sum_{i \in \mathcal{Q}_{\text{high}}} y_{jil} c_{jil}(t) \quad (27)$$

where $c_{jil}(t) = w_T T_{\text{leg}} / T_{\text{max}} + w_E E_{\text{leg}} / E_{\text{max}} - w_U V_i(t) / K$ per survivor normalized by K (so $\sum_{i \in \pi_j} u_i c_i / K \in [0, 1]$ tour-level), precomputed via oracle for position l , and $C(\pi_j) = w_T T(\pi_j) / T_{\text{max}} + w_E E(\pi_j) / E_{\text{max}} - w_U (\sum_{i \in \pi_j} u_i c_i) / K = \sum_{i,l} y_{jil} c_{jil}$. The single-pair C_{ij} is $k = 1$ where $y_{ji1} = 1$, where $T_{\text{leg}}(i), E_{\text{leg}}(i)$ include travel to i plus 15s hover. If $\mathcal{Q}_{\text{high}} = \emptyset$ or all $C(\pi_j) = \infty$, the optimum is $\pi_j = \emptyset$ (idle).

B. State Update for Tours

Dispatching $\pi_j = (i_1, \dots, i_k)$ updates: $\mathcal{Q}(t^+) = \mathcal{Q}(t^-) \setminus \{i_1, \dots, i_k\}$, $B_j(t^+) = B_j(t^-) - E(\pi_j)$, $I_j(t^+) = I_j(t^-) \setminus \bigcup_{i \in \pi_j} \mathcal{R}_i$, $m(t^+) = m(t^-) - \sum_{i \in \pi_j} \text{weight}(\mathcal{R}_i)$, $\text{status}_j \leftarrow \text{EnRoute}$ until return to p_j^{base} for reload.

IX. STATE TRANSITION FUNCTIONS

A. New Detection

$$\mathcal{S}(t^+) = \mathcal{S}(t^-) \cup \{S_{\text{new}}\} \quad (28)$$

$$\mathcal{Q}(t^+) = \mathcal{Q}(t^-) \cup \{S_{\text{new}}\} \quad (29)$$

B. Tour Dispatch

Dispatching tour $\pi_j = (i_1, \dots, i_k)$:

$$\mathcal{Q}(t^+) = \mathcal{Q}(t^-) \setminus \{i_1, \dots, i_k\} \quad (30)$$

$$B_j(t^+) = B_j(t^-) - E(\pi_j) \quad (31)$$

$$I_j^q(t^+) = I_j^q(t^-) - \sum_{i \in \pi_j} r_i^q \quad \forall q \quad (32)$$

$$\text{status}_j \leftarrow \text{EnRoute} \quad (33)$$

with $m_j(t^+) = \sum_q I_j^q(t^+) w_q$ derived.

C. Tour Completes (Return to Base)

Upon return to p_j^{base} :

$$\text{status}_j \leftarrow \text{Idle} \quad (34)$$

$$p_j \leftarrow p_j^{\text{base}} \quad (35)$$

$$I_j \leftarrow \text{reload}(K) \quad (36)$$

$$m \leftarrow \sum_{\text{item} \in I_j} \text{weight}(\text{item}) \quad (37)$$

$$A_j \leftarrow 1 \quad (38)$$

D. Expiration and Fairness

TABLE IV
EXPIRY/FAIRNESS PARAMETERS

Parameter	Value	Meaning
$u_{\min}$	0.05	Min urgency threshold ($u < u_{\min} \Rightarrow \text{expire}$)
T_{fair}	15 min	Fairness wait for $\sigma_i > 0.8$ (promote to top of $\mathcal{Q}_{\text{high}}$)

Expire: if $u_i(t) < u_{\min} \Rightarrow \mathcal{Q} \leftarrow \mathcal{Q} \setminus \{S_i\}$. Fairness: if $\sigma_i > 0.8 \wedge (t - t_i^{\text{det}}) > T_{\text{fair}} \Rightarrow \text{promote } S_i \text{ to top of } \mathcal{Q}_{\text{high}}$.

X. SCOPE AND LIMITATIONS

A. Core Model Includes

- Survivor detection as exogenous input
- Responder UAV assignment and routing
- Battery-aware feasibility
- Time-dependent urgency
- Online, event-triggered decision making

B. Excluded (Future Work)

- Dynamic aftershock obstacles
- UAV-to-UAV collision avoidance
- Communication relay UAVs
- Hot-swap battery stations
- Detection uncertainty verification policies
- Recourse/reassignment after dispatch
- High-fidelity nonlinear energy/power model (mass, aerodynamic drag, wind, acceleration-aware)

XI. CONCLUSION

This document presents a complete mathematical formulation of the Proactive Golden-Hour Bridge Model (PGBM) for post-earthquake UAV-assisted response. The model captures the essential dynamics of the system: survivor states with time-dependent urgency, Responder UAV states with battery constraints, and the online nature of detection arrivals and resource availability. The formulation separates feasibility (battery and routing constraints) from optimization (assignment and path selection), providing a clear foundation for algorithm design and simulation. The coupling between assignment and routing is explicitly defined through the tour cost function $C(\pi_j)$ (with $C_{ij} = C(\pi_j = (i))$ as the $k = 1$ special case), which depends on travel time and energy per leg and urgency per survivor.